Exceptional parabolics from their common normalizers

Human Compression Project

Abstract

A fixed small intersection has at most two exceptional overgroups of the required type. Counting those overgroups and then forgetting their order recovers the exceptional-family rows of the Part II classification. Dihedral second parabolics are uniquely specified by their rotations; mixed A_5, S_4 pairs are specified by an A_4 normalizer orbit. The same arguments cover both conjugacy classes and their fusion, without a numerical group census.

1 Scope and the two-class argument

Let $G = \mathrm{PSL}(2, q)$ with q odd. For $M \cong S_4$ assume $q = p$ is prime and $q \equiv \pm 1 \pmod 8$. For $M \cong A_5$ assume either $q = p \equiv \pm 1 \pmod 5$ with p odd prime, or $q = p^2 \equiv -1 \pmod 5$ with p odd prime. In these cases M is maximal, has two G-classes, and these classes are fused by $E = \mathrm{PGL}(2, q)$. These are the parameter ranges of the fixed Part II source [1]. The classical subgroup facts, as well as the small subgroup lattices of A_5 and S_4, are retained premises.

The common two-transitive overgroup lemma is a separate explicit interface for exhaustivity. It leaves the following possibilities:

M	B	possible other parabolic
A_5	D_{10}	D_{20}, D_{30}, A_5
A_5	A_4	S_4, A_5
S_4	D_6	D_{12}, D_{18}, S_4
S_4	D_8	D_{16}, D_{24}, S_4
S_4	A_4	A_5

A second S_4 over the normal subgroup A_4 would have to equal its unique ambient normalizer and is not a distinct parabolic. The lists are subject to the actual ambient divisibility conditions. The common lemma is proved in the separately reviewed manuscript *Two-transitive coset actions in dimension two*. All counting arguments below are independent of that exhaustivity step.

Lemma 1. *Suppose M has two ambient conjugacy classes exchanged by E, contains a single internal class of subgroups of type B, and $\mathrm{N}_M(B) = B$. Assume M is self-normalizing and B is maximal in M. Suppose also that the B subgroups form one E-class.*

1. *If B has two G-classes and $[\mathrm{N}_G(B) : B] = 2$, then there are exactly two copies of M above each B, both in the same G-class.*

2. *If B has one G-class and $\mathrm{N}_G(B) = B$, then there are exactly two copies of M above it, one in each G-class.*

Consequently unordered pairs of distinct copies of M meeting in B have respectively two or one conjugacy classes, and one class under E in either case.

Proof. For a fixed ambient class of M containing B, its members above B form a transitive $\mathrm{N}_G(B)$-set, by adjusting a transporter inside the target M. Their stabilizer is B. In the first case, each M class contains only one B class. The outer interchange of the two classes of each type forces the two M classes to contain different B classes. Hence the fixed B has exactly two overgroups in one class and none in the other. In the second case, each of the two M classes contributes exactly one member.

In both cases there is a unique unordered pair above B, and its intersection is exactly B by maximality of B in M. The intersection recovers B from the pair. Thus conjugacy and E classes of unordered pairs are exactly those of B itself. Distinct maximal parabolics generate G. Their prescribed local actions are two-transitive, so these pairs give the required geometries. □

2 Dihedral intersections

Let $B = D_{2d}$ with $d = 3, 4$ or 5. Its characteristic rotation subgroup lies in a unique torus of G, of order $n = (q - \epsilon)/2$, where $\epsilon = \pm 1$ and $d \mid n$. Write $u = n/d$. In the torus normalizer of E choose

$$\langle R, S \mid R^{2n} = S^2 = 1,\ SRS = R^{-1} \rangle, \qquad \mathrm{N}_G(C_d) = \langle R^2, S \rangle, \qquad B = \langle R^{2u}, S \rangle.$$

All such B form one E-class: the rotations are fixed by the torus and the possible reflections in G differ by even powers of R, which are conjugate under its full cyclic group. A rotation R^k normalizes B exactly when $u \mid k$. Therefore $\mathrm{N}_E(B) = \langle R^u, S \rangle \cong D_{4d}$. Its intersection with G is all of D_{4d} if u is even, and is B if u is odd. The E-class splits into two G-classes exactly when its normalizer lies in G, hence exactly when u is even.

Theorem 2. *Let (M, B) be one of*

$$(A_5, D_{10}), \qquad (S_4, D_6), \qquad (S_4, D_8).$$

The equal-type configurations $(M, M; B)$ have two conjugacy classes if $u = (q - \epsilon)/(2d)$ is even and one if it is odd. Their E-class count is one. For $k = 2$ or 3, configurations $(M, D_{2kd}; B)$ exist exactly when $kd \mid n$, and then have two conjugacy classes and one E-class.

Proof. Each indicated B is maximal and self-normalizing in M, and forms one internal class there. The equal-type counts follow directly from Lemma 1 and the normalizer calculation above.

A dihedral overgroup $H = D_{2kd}$ must normalize the same rotation torus: its cyclic rotation subgroup contains C_d. If $kd \mid n$, that torus has a unique subgroup C_{kd}, and $H = \langle C_{kd}, B \rangle$ is the unique overgroup of the required type containing the fixed B. Conversely its existence requires that divisibility. For each ambient class of M there is one class of chains (M, B), and the unique H over B then gives one configuration. The two classes of M give two conjugacy classes, interchanged by E.

The intersection $M \cap H$ equals B: otherwise maximality of B in M would place M inside a dihedral group, impossible. Maximality of M then gives generation of G. The action of D_{4d} on cosets of D_{2d} has degree two; that of D_{6d} has degree three with quotient S_3. These are two-transitive. The M-side actions are their standard small actions. □

The parity and divisibility tests recover Table 1 rows 5–8 and 12–15, and Table 3 rows 1–8, with the parameter qualifications below. In Table 3, $6 \mid n$, $9 \mid n$, $8 \mid n$ and $12 \mid n$ are exactly the printed congruences for rows 1,2,5,6. The full graph groups for the S_4, S_4, D_8 and S_4, D_{16}, D_8 cases are proved in separate accepted manuscripts; these counts do not substitute for those group-identification arguments.

3 The intersection A_4

The classical subgroup normalizers give

$$\mathrm{N}_G(A_4) = \begin{cases} S_4, & q \equiv \pm 1 \pmod 8, \\ A_4, & q \equiv \pm 3 \pmod 8. \end{cases}$$

In the first case there are two classes of A_4, and in the second there is one; they form one E-class. Within A_5 the subgroups A_4 form one class and are maximal and self-normalizing.

Theorem 3. *The configurations $(A_5, A_5; A_4)$ have two conjugacy classes when $q \equiv \pm 1 \pmod 8$ and one when $q \equiv \pm 3 \pmod 8$. In both cases they have one E-class. Configurations $(A_5, S_4; A_4)$ exist in the first case only, and have two conjugacy classes and one E-class.*

Proof. The equal-type assertion is Lemma 1. For the mixed pair, a fixed $B = A_4$ has the unique S_4 overgroup $N = \mathrm{N}_G(B)$. The two A_5 classes contain different A_4 classes, since the outer coset interchanges each pair of classes. Thus the fixed B belongs to only one class of A_5. Its members above B form a transitive N-set with stabilizer B, so there are two of them, interchanged by N. Hence there is one mixed configuration for each of the two B classes, and the outer automorphism interchanges them. The intersection is B; local indices are five and two. At least the A_5 parabolic is maximal, so the graph is connected. For $q \equiv \pm 3 \pmod 8$, the ambient group contains no S_4, so the mixed configuration does not exist. □

These statements recover Table 1 rows 9–11 and 16–18, and Table 3 row 9. The mixed $q = 9$ case is also Table 6 row 2; its identical normalizer argument is used in the subfield manuscript. For these families there is already one class under E, so adjoining field automorphisms cannot change the full automorphism count.

4 Empty parameter rows and reuse

In the square-field part of Table 1, the hypotheses impose $q = p^2 \equiv -1 \pmod 5$ with p odd. Every odd square is one modulo eight; combining this with $q \equiv 4 \pmod 5$ gives

$$q \equiv 9 \pmod{40}.$$

Consequently rows 12–14 and 18 have no admissible parameters: row 12 requires $q \equiv -1 \pmod{20}$; row 14 requires $(q+1)/10$ even; row 18 requires $q \equiv 29$ or $19 \pmod{40}$. Row 13 requires $q \equiv -1 \pmod{30}$, impossible since a square is never two modulo three. These are empty conditional branches, not numerical counterexamples. Rows 15–17 are the actual square-field configurations. In rows 16–17 the printed alternative $q \equiv -1 \pmod{40}$ is likewise empty, leaving $q \equiv 9$.

The remaining cases of Table 1 are already accounted for elsewhere: rows 2–4 use the accepted scalar-orbit proof in characteristic two; row 1 uses the unique D_{30} overgroup in the subfield manuscript, and repeats Table 4 row 5. That row's prime-exponent condition reduces to $q = 16$. The characteristic-five odd-degree case supplies no geometry, by the subfield uniqueness and exclusion proof. Thus no A5-family branch is silently omitted.

For source attribution, these are the configurations and numerical claims of L063 Propositions 5.5 and 5.10, Table 1 and Table 3. The retained L064 appendix gives the S4 counting argument in Proposition 20; its proof is recovered by the same formulas. The source's advertised A5-count

appendix pages are not present in that fixed edition, so the published A5-family proof is joined directly. The classical subgroup premises and the original constructions remain attributed. The common normalizer proof and explicit unordered-pair argument are our derivation. No additional principal refutation, conjecture answer, Lean verification or compression measurement is claimed. Whole-original completion still requires the source ledgers and the other shared proof components.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Table 1, Table 3 and Propositions 5.5, 5.10.

[2] The same authors, *Part II (Appendix)*, retained 21-page edition, Proposition 20.